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Chapter 7 — Structural and Computational Biology

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 7.1 to 7.4. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 7 carries no MCQ this year, but its concepts support short and long questions in other chapters. These notes cover the full textbook chapter.

TOPIC 7.1 — APPLICATIONS OF STRUCTURAL BIOLOGY

Textbook topic 7.1 · printed pages 167–169

Q.1 Describe the applications of structural biology in medical research.

Board long question · 8 marks

Structural biology deals with the study of three-dimensional structures of macromolecules such as proteins and nucleic acids at atomic level. It provides detailed information about structure, function, dynamics and interactions with ligands and other macromolecules.

1. **Determining active sites and domains:** 3D structures reveal the location, shape and environment of active sites and domains. For example, studies of HIV-1 reverse transcriptase identified its polymerase and RNase H domains, helping design antiviral drugs.
2. **Identifying drug targets:** Structural biology helps find specific spots on disease-causing proteins where drugs can bind. For example, the SARS-CoV-2 spike protein was identified as a drug target for vaccines and medicines.
3. **Identifying host-pathogen interactions:** It shows how pathogens attach to and enter host cells. Studying the coronavirus spike protein and its human receptor helps design drugs that block viral entry.
4. **Identifying protein misfolding:** Many diseases such as cystic fibrosis, Parkinson's and Alzheimer's arise from incorrect protein folding. Structural biology helps understand folding pathways and how misfolding leads to disease.

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Important Conceptual Short Questions — Topic 7.1

S1. Why is determining the 3D structure of a protein important for drug design?

It reveals the exact shape of active sites and domains, allowing drugs to be designed to bind specifically and block protein function.

S2. How did structural biology help in understanding HIV-1 reverse transcriptase?

It identified the polymerase domain that synthesizes DNA and the RNase H domain that breaks down RNA in RNA-DNA hybrids, helping design antiviral drugs targeting these domains.

S3. How can structural biology block viral entry into host cells?

By revealing how viral surface proteins bind host receptors, it allows design of drugs that inhibit this interaction and prevent viral entry.

S4. Why is protein misfolding linked to disease?

Protein function depends on correct three-dimensional folding. Misfolding disrupts function and can lead to diseases such as cystic fibrosis, Parkinson's and Alzheimer's.

TOPIC 7.2 — X-RAY CRYSTALLOGRAPHY

Textbook topic 7.2 · printed pages 169–170

Q.2 Explain how X-ray crystallography is used to determine protein structure.

Board long question · 8 marks

X-ray crystallography was developed in 1912 by **William Henry Bragg and William Lawrence Bragg**, who received the 1915 Nobel Prize in Physics. In this technique, an X-ray beam strikes a crystal, and atoms and molecules diffract the beam in specific directions. From the angles and intensities of diffracted beams, a 3D picture of electron density is produced and used to create the 3D structure of the molecule.

1. **Protein crystallization:** A purified protein is turned into a solid crystal with a regular, repeating pattern. This requires careful control of temperature, pH and salt concentration.
2. **Production of a diffraction pattern:** The crystal is mounted and bombarded with X-rays at various angles. Atoms diffract the X-rays, creating a pattern of spots on detectors.
3. **Creating a density map:** The angles and intensities of spots are used to make an electron density map.
4. **Determination of protein structure:** Computational programs analyse the data and transform it into the 3D structure of the protein.

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Important Conceptual Short Questions — Topic 7.2

S1. Why is crystallization necessary in X-ray crystallography?

Crystals arrange protein molecules in a regular, repeating pattern, which is needed to produce a clear diffraction pattern.

S2. How does a diffraction pattern reveal protein structure?

The angles and intensities of diffracted X-rays contain information about atom arrangement, which is used to build an electron density map and then a 3D model.

S3. Why are computational programs needed in X-ray crystallography?

They mathematically transform diffraction data into an electron density map and finally into the three-dimensional structure of the protein.

TOPIC 7.3 — COMPUTATIONAL BIOLOGY

Textbook topic 7.3 · printed pages 170–173

Q.3 Describe computational biology, its major areas, applications and key databases.

Board long question · 8 marks

Computational biology is an interdisciplinary field that uses computational techniques and tools to solve biological problems. It integrates biology, computer science, mathematics and statistics to analyse and interpret biological data, handle large datasets, uncover patterns and generate predictive models.

Major areas

- **Genomics:** Study of genomes, including sequencing, assembling and analysing function and structure of DNA.
- **Proteomics:** Large-scale study of proteins, their structures, functions and interactions.
- **Bioinformatics:** Application of computer technology to store, retrieve and analyse DNA, RNA and protein sequences.

Applications

- **Drug discovery:** Identifies drug targets and simulates drug effects.
- **Genetic research:** Identifies genetic variations associated with disease and supports personalized medicine.
- **Evolutionary biology:** Compares genetic information across species to reconstruct evolutionary relationships.

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Key databases and algorithms

Resource	Use
GenBank	Public database of nucleotide sequences and annotations
Protein Data Bank (PDB)	3D structural data of proteins and nucleic acids
Ensembl	Genome browser for genome sequences, gene models and comparative genomics
BLAST	Compares sequences to identify homologs, predict function and study evolution
FASTA	Sequence alignment tool to find regions of similarity

Important Conceptual Short Questions — Topic 7.3

S1. Why is computational biology essential for modern biological research?

It handles large datasets, uncovers hidden patterns and generates predictive models that would be impossible to analyse manually.

S2. How does genomics differ from proteomics?

Genomics studies the complete set of DNA and its function, while proteomics studies the complete set of proteins, their structures and functions.

S3. How does computational biology accelerate drug discovery?

It identifies potential drug targets and predicts how drugs interact with proteins and other molecules, reducing time and experimental cost.

S4. Why are databases such as GenBank and PDB important?

They store and provide access to sequence and structural data, enabling comparison, functional prediction and drug design.

S5. How does BLAST help in studying unknown genes?

It compares the unknown sequence with known sequences, identifying homologs and suggesting possible function and evolutionary relationships.

TOPIC 7.4 — SEQUENCE HOMOLOGY

Textbook topic 7.4 · printed pages 173–174

Q.4 Explain sequence homology and structural homology. How do they help in understanding evolution and protein function?

Board long question · 8 marks

Sequence homology refers to similarity between DNA, RNA or protein sequences due to shared ancestry. Homologous sequences evolve from a common ancestral sequence.

Type	Origin	Function
Orthologs	Sequences in different species that originated from a common ancestral gene during speciation	Often retain the same function across species
Paralogs	Sequences within the same species that originated from gene duplication	Can evolve new functions

Sequence homology provides insight into evolutionary relationships. By comparing homologous sequences, scientists can infer evolutionary history and divergence. If an unknown gene or protein is homologous to one with known function, it is likely to have a similar function. Homologous genes in model organisms help understand human diseases.

Structural homology refers to similarity in three-dimensional structures of proteins or macromolecules due to shared ancestry. Proteins with similar structures often perform similar functions even if their sequences are not highly

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similar. Structural homology helps predict function of newly discovered proteins, design drugs and understand evolution of protein function.

Important Conceptual Short Questions — Topic 7.4

S1. How can sequence homology predict the function of an unknown gene?

If the unknown gene is homologous to a gene with known function, it likely has a similar function because they share a common ancestor.

S2. Differentiate between orthologs and paralogs.

Orthologs are homologous sequences in different species that arose during speciation and often retain the same function. Paralogs arise by gene duplication within a species and may evolve new functions.

S3. Why can proteins with low sequence similarity still have similar functions?

Their three-dimensional structures may be homologous, and structure determines function more directly than sequence.

S4. How does structural homology help in drug design?

Drugs are designed to interact with specific protein structures. Understanding structural relationships helps design more effective drugs.

S5. How do homologous genes in model organisms help study human diseases?

They allow scientists to study disease mechanisms in model organisms and apply the insights to human diseases and treatments.

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